Full-Stack Setup Master Guide

Google Cloud OAuth & Resend SMTP (with cPanel DNS & Supabase Auth)

> ***Purpose:*** *This document provides an end-to-end, reusable blueprint for configuring* ***Google Authentication*** *via Google Cloud Console and* ***Transactional Email / Password Reset*** *via Resend with cPanel DNS in any web application powered by Supabase. Keep this guide as a reference for all current and future projects.*

Part 1: Google Cloud Console OAuth Setup (Start to End)

Google OAuth allows users to authenticate in one click using their Google accounts. The flow works by connecting your Google Cloud project to your Supabase Auth provider.

Step 1.1: Create Project & Configure OAuth Consent Screen

- Open the **Google Cloud Console**: https://console.cloud.google.com/
- Click the project dropdown at the top and click **New Project**. Name it your project name (e.g., *MyStore Production*).
- In the left navigation menu, go to **APIs & Services** → **OAuth consent screen**.
- Select User Type: **External** → Click **Create**.
- Fill in the required fields:
  - **App name:** Your Application Display Name (e.g. *Jog & Joy*)
  - **User support email:** Your admin email address
  - **Developer contact information:** Your admin email address
- Click **Save and Continue** through the **Scopes** screen (default scopes *email, profile, openid* are already selected).
- On the **Test users** screen, add your own Google email address while the app is in 'Testing' status. Click **Save and Continue**.

Step 1.2: Create OAuth 2.0 Client Credentials

- In the left menu, click **Credentials** → Click **+ CREATE CREDENTIALS** at the top → Select **OAuth client ID**.
- **Application type:** Select **Web application**.
- **Name:** Enter *Web Client*.
- **Authorized JavaScript origins:** Add your local and production domains:
  1) http://localhost:5173
  2) https://yourdomain.com
- **Authorized redirect URIs (Crucial):** Supabase uses a standardized callback URL format for every project:

```
https://<YOUR_SUPABASE_PROJECT_REF>.supabase.co/auth/v1/callback
```

- Click **Create**. A dialog will show your **Client ID** and **Client Secret**. Copy both.

Step 1.3: Enable Google in Supabase Dashboard

- Go to your **Supabase Dashboard** → Click **Authentication** → **Sign In / Providers**.
- Locate **Google** in the provider list and click to expand it.
- Toggle **Enable Sign in with Google** to **ON**.
- Paste your **Client ID** and **Client Secret** from Google Cloud Console.
- Click **Save**.

Step 1.4: Frontend Implementation Code

Part 2: Resend Custom SMTP & cPanel DNS Setup (Start to End)

Resend provides world-class email deliverability. To send emails from your own domain (e.g. *noreply@yourdomain.com*), you must verify domain ownership by adding DNS records in cPanel.

Step 2.1: Add Domain in Resend Dashboard

- Create an account and log in at https://resend.com/
- In the left sidebar, click **Domains** → click **+ Add Domain**.
- Enter your root domain (e.g. jognjoy.com) → Select region (default: US East) → Click **Add**.
- Resend will display a table containing **DKIM**, **SPF**, and **DMARC** DNS records.

Step 2.2: Add DNS Records in cPanel Zone Editor

- Log in to your hosting **cPanel** dashboard → Search for **Zone Editor**.
- Next to your domain, click **Manage** → Click the blue **+ Add Record** button.

Record	Type	Zone Name (in cPanel)	Value / Destination
DKIM	TXT	resend._domainkey cPanel appends .yourdomain.com.	Paste long string from Resend: k=rsa; p=MIGf...
SPF	TXT	bounces cPanel appends .yourdomain.com.	v=spf1 include:amazonses.com ~all
MX (Bounce)	MX	bounces (Priority: 10)	feedback-smtp.us-east-1.amazonses.com
DMARC	TXT	_dmarc cPanel appends .yourdomain.com.	v=DMARC1; p=none;

- Return to Resend and click **Verify Records**. Statuses will turn from *Pending* to green *Verified* (typically takes 5 to 15 minutes).

Step 2.3: Generate Resend API Key

- In Resend left menu, click **API Keys** → Click **Create API Key**.
- Name: Supabase SMTP → Permission: **Full access** → Click **Add**.
- Copy the API key starting with re_.... You will use this as your SMTP password.

Step 2.4: Connect Resend SMTP inside Supabase

- In Supabase Dashboard, click **Authentication** → under **NOTIFICATIONS** click **Emails**.
- Toggle **Enable custom SMTP** to **ON**.

Supabase Field	Exact Value to Enter	Explanation / Warning
Sender email address	noreply@yourdomain.com	Must be on your verified domain. **Do not use @gmail.com!**

Sender name	Your Store Name	Shown in the recipient's inbox (e.g. *Jog & Joy Kids*).
Host	smtp.resend.com	Official Resend SMTP host.
Port number	465	Port 465 (SSL/TLS) or 587 (STARTTLS).
Username	**resend**	Must be literally resend with a 'd' (not 'resent').
Password	re_xxxxxxxxxxxxxxxxxxxx	Your Resend API Key.

- Click **Save changes**.

Part 3: Password Reset Flow & Common Pitfalls Checklist

The password reset architecture operates across three distinct touchpoints: User Request → Supabase Resend Dispatch → In-App Password Update.

Step 3.1: Supabase Redirect URLs Configuration

- In Supabase Dashboard → **Authentication** → **URL Configuration**.
- **Site URL:** Set to your primary application URL (e.g. http://localhost:5173 or https://yourdomain.com).
- **Redirect URLs:** Add both wildcards and exact routes:
  - http://localhost:5173/**
    - http://localhost:5173/reset-password
    - https://yourdomain.com/reset-password
- Click **Save**.

Step 3.2: Complete Code Flow Diagram

```
// 1. Customer triggers password recovery in login modal:
await supabase.auth.resetPasswordForEmail(email, {
  redirectTo: `${window.location.origin}/reset-password`
});

// 2. Customer clicks link in email -> Opens /reset-password
// 3. Customer submits new password on ResetPassword.jsx:
const { data, error } = await supabase.auth.updateUser({
  password: newPassword
});
if (!error) {
  // Password successfully updated in Supabase!
}
```

Step 3.3: Troubleshooting & Verification Checklist

Symptom / Error	Root Cause	Immediate Fix
Auth 535 / Invalid Login	Typo in Resend username (e.g. *resent* with 't').	Change username to **resend** (with a 'd').
Cannot send from @gmail.com	Resend blocks sending from free Gmail addresses.	Use a sender from your verified domain (e.g. noreply@yourdomain.com).
redirect_uri_mismatch (Google)	Redirect URI in Google Cloud does not match Supabase callback.	Ensure https://<project>.supabase.co/auth/v1/callback is added in Google Console.
Reset link opens home page	Missing /reset-password in Supabase Redirect URLs.	Add http://localhost:5173/reset-password to Supabase URL Configuration.
DNS records still 'Pending'	DNS cache or typos in Zone Editor.	Verify record name in cPanel doesn't double-append domain (e.g. _dmarc.domain.com.domain.com).